ClauseLens - Product Requirements Document (PRD)

Project: ClauseLens: Private Legal Document Triage Agent **Event:** ZeroDay OSS Hacktoberfest 2026, IIIT Dharwad (Track PS2)
Version: 1.0 | **Deadline:** today, 8:20 PM

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1. Overview

ClauseLens is an open-source, fully on-device assistant that helps ordinary people understand legal documents such as rental agreements and employment contracts. It reads the document locally with a Gemma edge model, explains clauses in plain language, points to the exact passage behind every statement, separates what the text says from possible interpretations, admits when the document is silent or ambiguous, and recommends a qualified legal professional when needed.

It is **not a lawyer**. It supports preliminary understanding and issue spotting only. It never decides legal validity or enforceability.

2. Problem

People sign rentals, jobs and service contracts without understanding them. Professional help for every first question is impractical. Generic chatbots may invent clauses, answer from general knowledge, sound certain, and require uploading private documents to a cloud service. For legal papers, that is both unsafe and a privacy risk.

3. Goals and non-goals

Goals

1. Explain a document's key clauses, obligations, rights, deadlines, penalties and termination terms in plain language.
2. Ground every claim in a **verifiable passage** of the source document.
3. Distinguish explicit statements from interpretations and concerns.
4. Flag ambiguity, missing information and jurisdiction-dependent issues; say "the document does not say" when true.
5. Refuse verdicts ("is this enforceable?") and give specific escalation advice.
6. Keep all document content on the device, with a reproducible way to demonstrate it.
7. Be reusable on any in-scope document, not one hard-coded sample.

Non-goals (today)

- Legal advice, validity or enforceability decisions, outcome predictions
- Cloud processing of any kind, accounts, sync, telemetry
- OCR for scanned documents, multilingual output, two-version comparison (stretch)
- Court judgments or statute research (reference datasets are optional)
- Mobile apps, multi-user support

4. Scope

English-language **residential rental agreements** and **employment contracts**, as PDFs with a text layer or plain text. The README must state this honestly and the tests stay inside it.

5. Personas

Persona	Need	Example question
Tenant / first-time renter	Understand deposit, rent increases, notice, penalties	"How much notice do I need to give to leave?"
New employee	Understand notice period, non-compete, probation	"Can they terminate me without notice?"
Student / non-lawyer	Learn legal terms in plain words	"What does indemnify mean here?"
Judge / reviewer	Verify grounding and privacy	Clicks a citation, checks the network log

Primary demo persona: a tenant reading a rental agreement on a laptop with no internet.

6. User stories

ID	As a...	I want to...	So that...
U1	user	open a PDF or text file locally	my document stays on my device
U2	user	read a short plain-language summary	I get the gist fast
U3	user	see a list of key obligations, rights, dates, penalties, termination terms	I know what matters
U4	user	click any claim and see the exact source passage highlighted	I can verify it
U5	user	see what the text states separately from possible interpretations	I do not mistake a guess for a fact
U6	user	see how uncertain a point is and why	I know where to be careful
U7	user	be told when the document does not cover something	I am not given invented terms
U8	user	ask follow-up questions about the document	I can dig into my own concerns
U9	user	get a clear limit and a specific referral when I ask for a legal verdict	I know when to see a professional
U10	user	see a privacy panel listing every network call	I can trust it with sensitive papers
U11	user	delete the document and its data	I stay in control
U12	judge	run it offline and see zero outbound connections	I can confirm privacy
U13	judge	feed an unscripted or hostile document	I can test robustness

7. Functional requirements

ID	Requirement	Priority	Ref
FR1	Upload/open PDF or plain text, locally	Must	L1
FR2	Extract text and segment into passages with page and offsets	Must	L2
FR3	Gemma produces findings (category, plain explanation, kind, uncertainty, citations)	Must	L2, L3, L4, L10
FR4	Every finding cites passage IDs and a verbatim quote	Must	L5
FR5	System verifies quotes against the source; unverified claims are dropped or hidden	Must	L5
FR6	Each finding labelled `explicit`, `interpretation` or `concern`	Must	L6
FR7	Uncertainty level and reasons per finding; jurisdiction flag where relevant	Must	L7
FR8	"Document is silent" list for expected-but-missing clauses	Must	L7
FR9	Q&A over the document with cited answers	Must	L3, L5
FR10	Router detects legal-conclusion and off-topic questions and answers with a limit plus escalation	Must	L8
FR11	Persistent not-legal-advice disclaimer in UI	Must	PS2 boundary
FR12	Offline operation after setup; in-app network guard	Must	L9
FR13	Privacy tab with declared network calls and run-time guard status	Must	L9
FR14	Delete document, temp files and in-memory data	Must	Privacy
FR15	Prompt-injection handling: document text treated as data, suspicious passages flagged	Must	Criterion 6

ID	Requirement	Priority	Ref
FR16	Graceful errors: empty file, no text layer, oversize, wrong type, model timeout	Must	Criterion 6
FR17	Key dates and amounts table (structured extraction)	Should	L4
FR18	Benchmarks: memory, speed, load time on demo hardware	Must (criterion 5)	S3
FR19	Export findings as text or Markdown, with disclaimer	Could	UX
FR20	Local OCR for scans	Could	OPT
FR21	Two-version comparison	Could	OPT

8. Non-functional requirements

Area	Requirement
Privacy	No document content, derived text, questions or answers leave the device. No telemetry. No CDN assets. Logs contain no content
Grounding	0 displayed findings with unverifiable quotes
Safety	No verdicts on legality, validity or enforceability. Disclaimer always visible
Robustness	Handles malformed files, long documents, vague or conflicting wording, hostile text without crashing
Performance	First findings within about 2 minutes for a 10-page document on a typical laptop with E4B; show progress per stage
Reproducibility	Fresh clone plus README reproduces setup and demo; pinned dependencies; model acquisition documented
Accessibility	Keyboard use, visible focus, 4.5:1 contrast, ARIA labels, no meaning by colour alone
Openness	MIT licence; datasets and licences documented; no private or non-redistributable data committed

9. UX requirements (summary; detail in `FRONTEND_DESIGN.md`)

- Two-pane layout: document on the left, analysis on the right, linked by citations
- Visual difference between "The document says" and "Possible interpretation or concern"
- Clear uncertainty indicators and escalation panel
- Persistent disclaimer and "Local only" privacy badge
- Plain-language, non-alarming wording for non-lawyers

10. MVP vs stretch

Tier	Items
MVP (must ship)	FR1-FR16, FR18; sample docs; offline proof; docs; adversarial tests; demo video
Stretch	FR17 key-dates table, FR19 export, dark mode
Cut first when late	Export, dark mode, key-dates polish, OCR, comparison
Never cut	Quote verification, silent-clause handling, refusal + escalation, disclaimer, offline proof, README, injection test

11. Success metrics

Metric	Target
Displayed findings with verified citations	100%
Fabricated citations reaching the UI	0
"Document silent" cases answered correctly on our set	4 of 4
Legal-verdict questions limited and escalated	3 of 3
Injection test documents: instruction ignored	3 of 3
Non-loopback connections during analysis	0
Clause-question accuracy on our set	80% or higher, reported honestly
Fresh-clone setup following README	Succeeds
Gates G1-G5 and PS2 gates	All pass

12. Alignment with PS2 scoring rubric

Criterion	Pts	How the product addresses it
1. Accuracy and usefulness	15	Category-tagged findings, plain explanations, eval set
2. Grounding and traceability	15	Verified verbatim quotes, passage highlight, silence handling
3. Uncertainty, refusal, escalation	15	`kind` labels, uncertainty reasons, router, specific escalation
4. On-device privacy	15	Loopback-only, netguard, offline run, netmon log, threat model
5. Gemma integration	10	Gemma does extraction and explanation locally; model docs and benchmarks
6. Robustness	12	Malformed, long, vague, adversarial inputs handled
7. Usability	8	Two-pane UI for non-lawyers, prioritised findings
8. Reusability and docs	10	Works on any in-scope document; stated scope; local-processing design documented

Final score = 0.70 x track + 0.30 x common rubric. **Cap: 40 if criterion 2, 3 or 4 is Poor.**

13. Assumptions, dependencies, risks

Assumptions: a team laptop can run Gemma E2B or E4B via Ollama; Gemma licence accepted; setup-time internet is available once.
Dependencies: Gemma weights, Ollama, Python 3.11, Node 20.

Risk	Impact	Mitigation
Gemma invents or misquotes a clause	Fabrication cap	Verbatim quote verification, drop failures, summary from verified findings only
Small model gives weak extraction	Low accuracy	Window the document, constrain categories, few-shot example, E4B if RAM allows
Gemma too slow	Poor demo	E2B, shorter windows, cache analysis, progress UI
Context limit on long documents	Missed clauses	Map-reduce over windows; BM25 for Q&A
Privacy claim cannot be proven	Cap	Offline recording, netmon log, code review notes, honest threat model
UI loads an external font or script	Privacy failure	Bundle assets, strict CSP, check in netmon
Output read as legal advice	Penalty	Disclaimer, router, wording rules, escalation
Time overrun	Incomplete submission	MVP-first scope and cut list
Committing private or third-party docs	Gate fail	Synthetic samples only; scan before submit

14. Wording rules (product voice)

- Say "The document says..." for explicit text and "This may mean..." or "Worth checking..." for interpretation.
- Never say "this is legal", "this is illegal", "enforceable", "you should sign" or "you will win".
- Always name the clause or passage and suggest the kind of professional (for example a tenant lawyer, an employment lawyer).

15. Open questions to settle today

1. Exact Gemma variant and Ollama tag, and has the licence been accepted?
2. Which laptop runs the final demo, and how much RAM does it have?
3. Rental, employment, or both for the demo? (Recommend both, with one sample each.)
4. How will the offline test be shown (airplane mode, firewall, or `unshare`/container)?

5. Submission portal and format from the organisers.